EXPERIMENT 4

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Semester: 5

Subject Name: ADBMS **Subject Code: 24CSP-333**

Experiment 4 -- To study and implement advanced SQL queries using Joins, Corelated Query's and Conditional statement.

Aim : To understand the use of correlated subqueries in SQL by developing queries that retrieve department-wise top three distinct salary records while effectively managing duplicate salary values through joins and aggregate functions.

Question :4 Part A (i):

You are given a table tbl_happiness that contains the ranking of happiness for different countries. Your task is to retrieve all the records from the table **while ensuring that "India" and "Sri Lanka" appear at the top, followed by the rest of the countries in their original ranking order.**

```
Select rankings, country
FROM (
 SELECT rankings, country, (
        CASE
                WHEN country = 'India' THEN 1
                WHEN country = 'Sri Lanka' THEN 2
                ELSE 3
        END
        ) as ranks
 FROM tbl_happiness
 ORDER BY ranks, rankings
)
```

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	rankings integer	country character varying (50)
1	126	India
2	128	Sri Lanka
3	1	Finland
4	2	Denmark
5	3	Iceland
6	4	Israel
7	5	Netherlands
8	6	Sweden
9	7	Norway

Question: Part A (ii)

Make all the rankings in descending order except **India** and **Sri Lanka** SQL

```
Select rankings, country
FROM (
 SELECT rankings, country, (
	CASE
		WHEN country = 'India' THEN 1
		WHEN country = 'Sri Lanka' THEN 2
		ELSE 3
	END
	) as ranks
 FROM tbl_happiness
 ORDER BY ranks ASC, rankings DESC
)
```

	rankings integer	country character varying (50)
1	126	India
2	128	Sri Lanka
3	7	Norway
4	6	Sweden
5	5	Netherlands
6	4	Israel
7	3	Iceland
8	2	Denmark
9	1	Finland

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Question: Part B

A company maintains employee information in a single table where each employee may report to another employee acting as their manager. Prepare a report that displays the details of every employee who supervises at least one other employee. The report should include the manager's employee ID, name, total number of direct reports, and the average age of those reporting employees. The average age should be rounded to the nearest whole number and the final result should be sorted by employee ID.

```
select m.employee_id, m.name ,count(*), sum(e.age)/count(*) as
average_age from employees as e JOIN employees as m on e.reports_to =
m.employee_id group by m.employee_id, m.name
```

	employee_id [PK] integer	name character varying (50)	count bigint	average_age bigint
1	9	Hercy	2	38

Learning Outcome

After completing this experiment, the learner will be able to:

1. Implement correlated subqueries for row-wise comparisons in relational databases.
2. Utilize aggregate functions and `DISTINCT` to identify the top three salary values within departments.
3. Integrate joins with filtering conditions to retrieve accurate information from multiple tables.
4. Develop optimized SQL queries for ranking, grouping, and managing duplicate records.